

A Novel Neural Network-Based Transistor Compact Model Including Self-Heating

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Abstract—We develop a SPICE-compatible neural network-based compact model to accurately capture the temperature dependence and self-heating effects in Field Effect Transistors (FETs). The model is based on artificial neural networks with no semi-empirical temperature equations. The transfer and activation functions are optimized to improve the accuracy of the model. A new temperature relaxation model is proposed, which allows training the model using ambient temperature data without iteratively extracting the self-heating parameters. The proposed method can simply generate the ambient and dynamic self-heating characteristics for circuit simulations. The model can accurately reproduce the current-voltage (IV), capacitance-voltage (CV), and transient characteristics of FETs across a broad temperature range with a speed advantage of up to 12X versus BSIM-CMG.

Index Terms—Compact model, neural network, machine learning, temperature, self-heating.

I. INTRODUCTION

THERMAL effects play a crucial role in determining device and circuit performance [1], [2], [3], [4]. Modern-day chips house billions of transistors, exhibiting a significant power density, operating at temperatures substantially higher than room temperature. Elevated device temperatures result in the degradation of mobility and subthreshold swing, shifts in threshold voltage, and reliability concerns. Self-heating (SH) effects contribute to a gradual increase in device temperature, thereby impacting circuit performance. With V_{DD} scaling stalled at around 0.7V and the projected rise in densities of transistors and 3D chiplet packaging, temperature and self-heating will be increasingly important to the accuracy of circuit simulation.

Conventional physics-based compact models [5], [6], [7], [8] start from modeling the temperature dependence of physical quantities such as mobility and bandgap. Then, a separate self-heating network is used to update the temperature in the device. While extracting the model parameters, it is

required to iteratively adjust the temperature and self-heating parameters. In addition to physics-based compact models, neural network (NN)/machine learning (ML)-based compact models follow the same approach to develop temperature models [9], [10], [11], [12], [13]. These models use device temperature as an input and either need to evaluate the SH subcircuit self-consistently with the NN model during the training [12], or extract/ remove the self-heating effects before the training [10], [11], [13]. These approaches require SH evaluation before or during the training, making the extraction procedure complicated.

This work proposes a new temperature and self-heating modeling approach with our NN transistor compact model [14], [15]. The proposed methodology starts with $I(T_0)/Q(T_0)$, not $I(T)/Q(T)$, where T is the device temperature and T_0 is the ambient temperature. $I(T_0)$ can simply come from DC measurements instead of pulse measurements. The model can be trained with $I(T_0)$ and extract or recover the self-heating-free characteristics and perform the dynamic self-heating simulation with a newly developed temperature relaxation model. The extracted characteristics can also be integrated with the standard SH model for circuit simulations.

II. MODEL

Our NN compact model is trained using DC-IV and CV characteristics including the SH effect. The model is enhanced from our previous work [14], [15] by adding T_0 as an input parameter and a new transfer and activation function to improve accuracy. We train our model to be a function of T_0 instead of T , such that no SH evaluation is required during training. Then, a new temperature relaxation (TR) model is used to recover the SH-free characteristics and model SH effect after the model is trained. V_{GS} , V_{DS} , L , W , EOT , and T_0 are the designated input parameters for our model. The outputs are the corresponding DC currents and charges at T_0 , with a new transfer function for I_D presented in (1), which preserves the V_{DS} sign dependence in I_D with a new parameter α to adjust the trained output y_1 for accuracy. The new activation function (2) is called the Inverse Square Root Unit (ISRU) function. ISRU is more computationally efficient than sigmoid and tanh implementations. The parameter β can also be tuned to improve the fitting accuracy. The loss function is similar to the one [15] using current (I) and charge (Q) and their derivatives as the loss. In the scenario where SH-free data is

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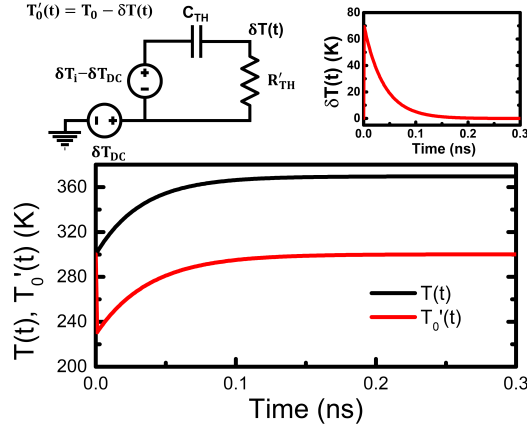


Fig. 1. The subcircuit of the temperature relaxation model and the simulated $T(t)$, $T'_0(t)$, and $\delta T(t)$ from the traditional self-heating model and the temperature relaxation model. A step function voltage is applied.

available, it can directly be trained using T as an input to get $I(T)$ and $C(T)$.

$$I_D = \tanh(\alpha V_{DS}) e^{y_1}, y_1 = \ln\left(\frac{I_D}{\tanh(\alpha V_{DS})}\right), \quad (1)$$

$$\text{ISRU}(x) = \frac{x}{\sqrt{1 + \beta x^2}}, \quad (2)$$

No semi-empirical temperature equations are used in the proposed model. Thus, the model is not restricted to certain predefined temperature dependencies.

To model the SH dynamic, we propose a new temperature relaxation (TR) model to map the device temperature to the corresponding ambient temperature. The traditional way to model the device temperature $T(t)$ is $T(t) = T_0 + \Delta T(t)$, where $\Delta T(t)$ is the change in temperature due to SH. We define a temperature $T'_0(t)$ such that $T_0 = T'_0(t) + \delta T(t)$ and $T'_0(t) = T_0 - \delta T(t)$ (3). $\delta T_i = R_{TH} I_D V_{DS}$ (4) is introduced, which is the $\delta T(t)$ to recover the SH-free characteristics. It is also the same as the steady-state $\Delta T(t)$ in the traditional model. However, the thermal resistance for the trained device (R_{TH}) and the device for IC design (R'_{TH}) can be different. $\Delta T(t)$ for the latter is $R'_{TH} I_D V_{DS}$ and the steady-state $\delta T(t) = \delta T_i - R'_{TH} I_D V_{DS}$ (4). The dynamic of $\delta T(t)$ is then modeled by (5), where $\tau = R'_{TH} C_{TH}$ is the thermal time constant and C_{TH} is the thermal capacitance. Fig. 1 shows the subcircuit for (4) and the simulation results. The trained DC NN model is integrated with the subcircuit in SPICE simulations. $T'_0(t)$ is self-consistently updated during the simulations and fed to the input of the NN model to replace T_0 . By mapping the input temperature to different $T'_0(t)$, the model tracks the corresponding T_0 and I_D determined by SH. In this case, $T'_0(t)$ decreases at $t = 0$ since no heat has accumulated yet, and $\delta T(t)$ is at its maximum. It eventually relaxes to the ambient condition to get the correct DC characteristics.

$$T'_0(t) = T_0 - \delta T(t) \quad (3)$$

$$\delta T_i = R_{TH} I_D V_{DS}, \delta T_{DC} = (R_{TH} - R'_{TH}) I_D V_{DS}, \quad (4)$$

$$d(\delta T_i - \delta T(t))/dt = (\delta T(t) - \delta T_{DC})/\tau, \quad (5)$$

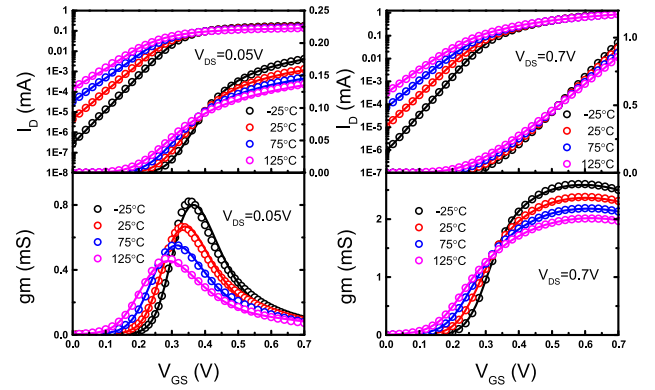


Fig. 2. Validation of transfer characteristics at different temperatures. The lines are the NN model, and the symbols are the BSIM-CMG data.

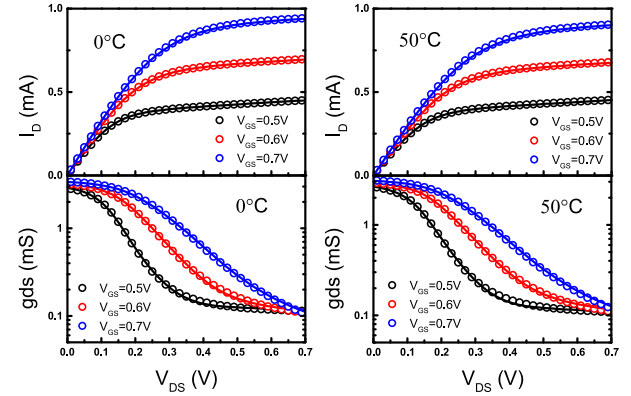


Fig. 3. Validation of output characteristics at different temperatures. The lines are the NN model, and the symbols are the BSIM-CMG data.

III. VALIDATIONS

Our model is trained using TensorFlow with the Adam optimizer and a learning rate of $1E-3$. Both IV and CV networks have two hidden layers with 30 and 20 neurons. It is then coded into Verilog-A for circuit simulations using HSPICE on an Intel Xeon Platinum 8260 workstation. A calibrated BSIM-CMG model card [15] is used to generate DC IV and CV data for training, validation, and testing. The model is trained from -50°C to 100°C . Fig. 2 & 3 show the validation of $I_D V_G$ and $I_D V_D$ at different temperatures. It accurately models the temperature dependence of I_D , g_m , and g_{ds} without using additional model equations. Fig. 4 shows the model fitting of C_{gg} , C_{gs} , C_{gd} , and C_{dd} . The model demonstrates excellent fitting results of these capacitances at a wide temperature range. Moreover, the 125°C IV and CV data is accurately modeled - showcasing the predictivity of the NN model. The simulations for training are performed under DC conditions with self-heating in effect, and the model can generate the characteristics without computing the self-heating subcircuit.

Then, to obtain the time-dependent self-heating characteristics, the temperature relaxation model and the subcircuit are used. In Fig. 5, we compare the NN model and the BSIM-CMG IV curves with the SH effect turned on and off. The proposed TR model can successfully recover the SH-free characteristics. We adjust R_{TH} , which changes $\delta T(0)$ to fit the SH-free data. T'_0 is decreased and the device can be mapped to the higher current values corresponding to the case without SH. Finally, to demonstrate the proposed subcircuit

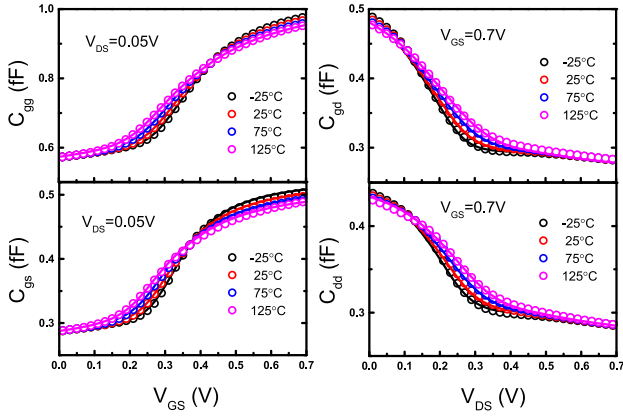


Fig. 4. Validation of CV characteristics at different temperatures. The lines are the NN model, and the symbols are the BSIM-CMG data.

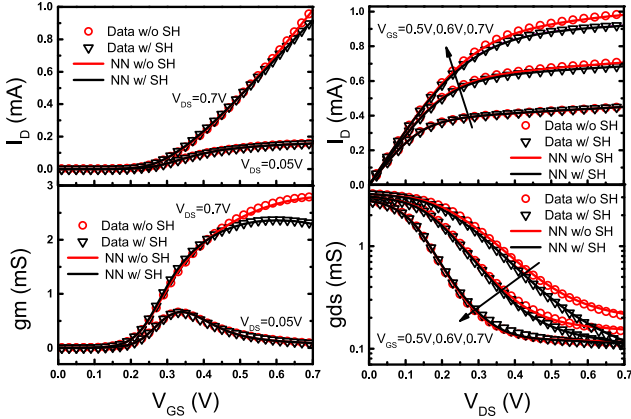


Fig. 5. Validation of the SH-free characteristics recovered by the temperature relaxation (TR) model.

for transient simulations, a pulse simulation is performed. Fig. 6 shows the pulse simulation results. A transistor is biased at $V_{GS} = 0.7V$ and V_{DS} pulses are applied with varying amplitudes. We show the simulation at 10 GHz, and the results generated by the NN model accurately match the output from BSIM-CMG. The C_{TH} is tuned so that the time constant of the NN model matches the testing data. R_{TH} is extracted from Fig. 5 which can also be obtained by fitting the peak current. With the TR model (red), the device current gradually decreases over time due to heat dissipation. Without the TR model (blue), the current will attain a steady state with no time delay. We can also use this model as an intermediate model to generate an SH-free $I(T)/Q(T)$ model (Fig. 5) and combine it with the conventional SH model (6) (black solid). By tuning R'_{TH} , the model can capture SH and steady-state characteristics from the training data (black dash).

$$\Delta T(t) + \tau_d \Delta T(t)/dt = R'_{TH} I_D V_{DS}, \quad (6)$$

Finally, to test the efficiency of our model, we ran a 10ns transient simulation on a s27 ISCAS89 circuit [16]. The runtime for the NN model was 16.86s, significantly faster than the 47.46s for BSIM-CMG. We further test the speed using a 1001-stage NAND oscillator (Fig. 7). With the TR model, the result shows a faster oscillation frequency compared to the case without TR, which matches our expectations. The simulation time is only 464.85s for the NN model with TR, similar to the case without TR (431.37s), and is about 12X

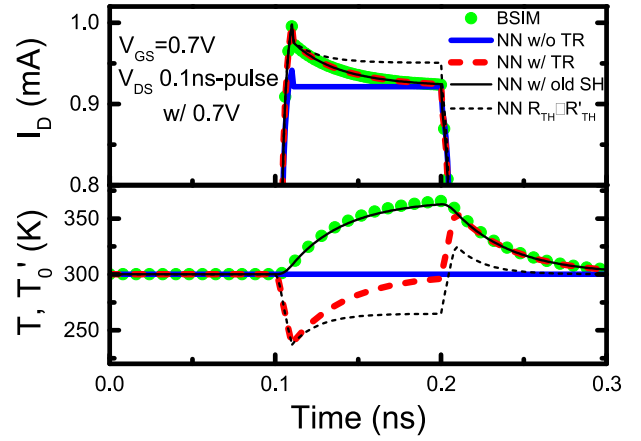


Fig. 6. Transient pulse simulations of BSIM-CMG, the NN model w/ and w/o TR model, the extracted SH-free NN with conventional SH model, and the case that $R_{TH} \neq R'_{TH}$.

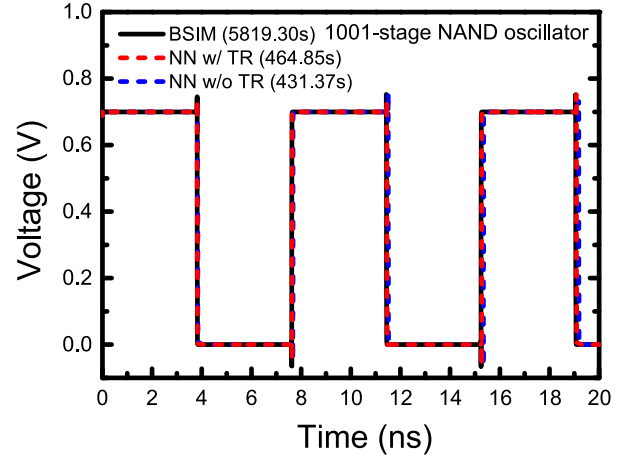


Fig. 7. 1001-stage NAND oscillator simulations of the NN model and BSIM-CMG.

faster than BSIM-CMG (5819.30s). These runtimes exhibit the accuracy and computational efficiency of the proposed NN and TR model.

IV. CONCLUSION

A temperature-dependent NN transistor compact model with self-heating is proposed. It requires no predefined semi-empirical temperature functions and can model transistors across a wide temperature range. A new temperature relaxation modeling approach is implemented. The NN model can be trained using just DC characteristics without iterations in training or eliminating the SH effect during measurements. The model automatically generates DC IV and CV with the SH effect. The self-heating dynamic is modeled using the proposed temperature relaxation subcircuit with excellent accuracy. The proposed model further offers the flexibility to use the conventional self-heating subcircuit on an extracted SH-free NN model. The proposed model is 12 times faster than BSIM-CMG when running a 1001-stage NAND oscillator.

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